Multiples, Factors & Primes

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Written by: Jamie Wood
Reviewed by: Dan Finlay
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Multiples

What are multiples?

- A **multiple** of a given integer is a number that can be formed by **multiplying** the integer by another positive integer
  - For example, 12 is a multiple of 3 because 12 = 3 × 4
- Multiples can be considered as the numbers in a times table
- However multiples go beyond times tables and continue forever
  - For example, the multiples of 3 are 3, 6, 9, 12, 15, ..., 300, ..., 3000, ..., 34 567 896, ...
- Every non-zero number has an **infinite** number of multiples
- A **common multiple** of two or more integers is a number that is a multiple of the integers
  - For example, 12 is a common multiple of 4 and 6 6
- **Even** numbers (2, 4, 6, 8, 10, ...) are multiples of 2
- **Odd** numbers (1, 3, 5, 7, 9, ...) are **not** multiples of 2
- Multiples can be **algebraic**
  - For example, the multiples of k would be $k, 2k, 3k, 4k, 5k. \ldots$

How do I find the multiples of a number?

- Starting with a particular value, multiples can be listed by counting up in steps of that particular value
  - e.g. the multiples of 7 start with 7, then counting up in 7's will give 14, 21, 28, 35 and so on
- Multiples form a sequence
  - e.g. 7, 14, 21, 28, 35, ...
- Questions may ask you to state the multiples of a value between certain numbers
  - e.g. the multiples of 7 between 10 and 40 are 14, 21, 28 and 35

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Answer:

$2, 4, 6, 8, 10$

(b) List the multiples of 5 between 12 and 37.

Answer:

$15, 20, 25, 30, 35$

Factors

What are factors?

Help me understand

- A **factor** of a given integer is a value that divides the given number exactly with **no remainder**
  - 6 is a factor of 18 because 18 ÷ 6 = 3
- Every integer **greater than 1** has **at least two factors**
  - The integer itself, and 1
- A **common factor** of two or more integers is a number that is a factor of the integers
  - For example, 3 is a common factor of both 21 and 18

How do I find factors?

- Finding all the factors of a particular value can be done by finding factor pairs
- For example when finding the factors of 18
  - 1 and 18 will be the first factor pair
  - Divide by 2, 3, 4 and so on to test if they are factors
    - 18 ÷ 2 = 9, so 9 and 2 are factors
    - 18 ÷ 3 = 6, so 6 and 3 are factors
    - 18 ÷ 4 = 4.5, so 4 is not a factor
    - 18 ÷ 5 = 3.6, so 5 is not a factor
    - 18 ÷ 6 would be next, but we have already found that 6 was a factor
    - So we have now found all the factors of 18: 1, 2, 3, 6, 9, and 18

How do I find factors without a calculator?

- Use a **divisibility test**
  - Some tests are easier to remember, and more useful, than others

Divisible by...	Test	Examples

		• Ends in a 3 134 is divisible by 2 • Ends in a 4
3	The **sum of the digits** is a 3, 6 or 9 • If the sum is bigger than 9 • Repeat the process with the digits in the sum	123 is divisible by 3 • 1+2+3=6 2574 is divisible by 3 • 2+5+7+4=18 • 1+8=9 134 is **not** divisible by 3 • 1+3+4=8
4	The **last two digits** can be **halved twice** to give a whole number	528 is divisible by 4 • 28 ÷ 2 = 14 • 14 ÷ 2 = 7 4274 is **not** divisible by 4 • 74 ÷ 2 = 37 • 37 ÷ 2 = 18.5
5	The **last digit** is 0 or 5	3025 is divisible by 5 • Ends in a 5 2719 is **not** divisible by 5 • Ends in a 9
8	The **last three digits** can be **halved three times** to give a whole number	2144 is divisible by 8 • 144 ÷ 2 = 72 • 72 ÷ 2 = 36 • 36 ÷ 2 = 18 4916 is **not** divisible by 8 • 916 ÷ 2 = 458 • 458 ÷ 2 = 229 • 229 ÷ 2 = 114.5

	than 9 • Repeat the process with the digits in the sum	• 4+2+3=9 2574 is divisible by 9 • 2+5+7+4=18 • 1+8=9 134 is **not** divisible by 9 • 1+3+4=8
10	The **last digit** is 0	2710 is divisible by 10 • Ends in a 0 3025 is **not** divisible by 10 • Ends in a 5

- Once you know that the number has a particular factor, you can divide by that factor to find the factor pair
- If a number is **not** divisible by an integer, then the number is also **not** divisible by **any multiple** of that integer
  - 119 is not divisible by 3
    - so 119 is not divisible by 6, 9, 12, etc, ...
- If a number is divisible by **two integers**, and the **only common factor** of those integers is **1**, then the number is **also divisible** by the **product** of the integers
  - 594 is divisible by 2 and 3
  - The only common factor of 2 and 3 is 1
    - so 594 is divisible by 2×3=6
- Instead of a divisibility test, you could use a formal written method to divide by a value
  - If the result is an integer; you have found a factor

Examiner Tips and Tricks

- On the calculator exam paper, use your calculator to test for divisibility
  - A factor pair will be found if the result of the calculation is an integer
- Being very familiar with times tables helps to reduce the need to use the divisibility tests

Worked Example

Find all the factors of 42.

1 42

42 ends in a 2 so 42 is divisible by 2

- 42 ÷ 2 = 21

2 21

4+2=6 so 42 is divisible by 3

- 42 ÷ 3 = 14

3 14

42 ÷ 2 = 21 and 21 ÷ 2 =10.5 so 42 is not divisible by 4

42 ends in a 2 so 42 is not divisible by 5

42 is divisible by 2 and 3 so 42 is divisible by 6

- 42 ÷ 6 = 7

6 7

The next number to test, 7, is already on the list, so the list is complete

Write the list of factors in order, being careful to not miss any out

The factors of 42 are 1, 2, 3, 6, 7, 14, 21 and 42

Prime numbers

What are prime numbers?

- A **prime** number is a number which has exactly two (**distinct**) **factors;** itself and 1
  - The first 10 prime numbers are
    - 2, 3, 5, 7, 11, 13, 17, 19, 23, 29
    - You should remember at least the first ten prime numbers
- 1 is **not** a prime number, there are a few reasons for this such as
  - by definition, prime numbers are integers greater than or equal to 2
  - 1 only has one factor
- 2 is the only even prime number
- If a number has any factors other than itself and 1, it is not a prime number
  - For example, 27 is often mistaken for a prime number

Worked Example

Show that 51 is **not** a prime number.

is not prime

51 is not even so is not divisible by 2
Next use the divisibility test for 3

5 + 1 = 6 and 6 is divisible by 3
therefore 51 is divisible by 3
51 ÷ 3 = 17

51 is not prime as it has more than two (distinct) factors

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